

Artificial Intelligence Cloud Innovation Center Discovery Workshop Readout



UI/UX:

Who are the users of this application?

- Community Health Officers (CHOs) and similar field-level personnel who attend virtual knowledge-sharing sessions
- Healthcare workers and doctors from tier 2 and tier 3 cities participating in specialized training programs
- Non-specialist doctors seeking to become more specialized in domains like TB, mental health, and other healthcare areas
- Frontline workers including farm extension workers (Krishi Mitras), teachers, and educators across various sectors
- Authenticated learners and participants enrolled in Echo's programs across healthcare, education, agriculture, livelihood, and other complex domains

Why are the users on this chatbot?

- **Post-session knowledge retention** - To recall and apply knowledge from virtual training sessions they attended
- **Quick access to approved content** - To get immediate answers from vetted, expert-approved documents within their specific program
- **Protocol and checklist access** - To quickly recall key protocols, definitions, and checklists relevant to their fieldwork
- **On-demand learning support** - To ask questions and get accurate responses when they need clarification during field work
- **Overcoming connectivity limitations** - To access essential information even with poor internet connectivity on low-end Android devices

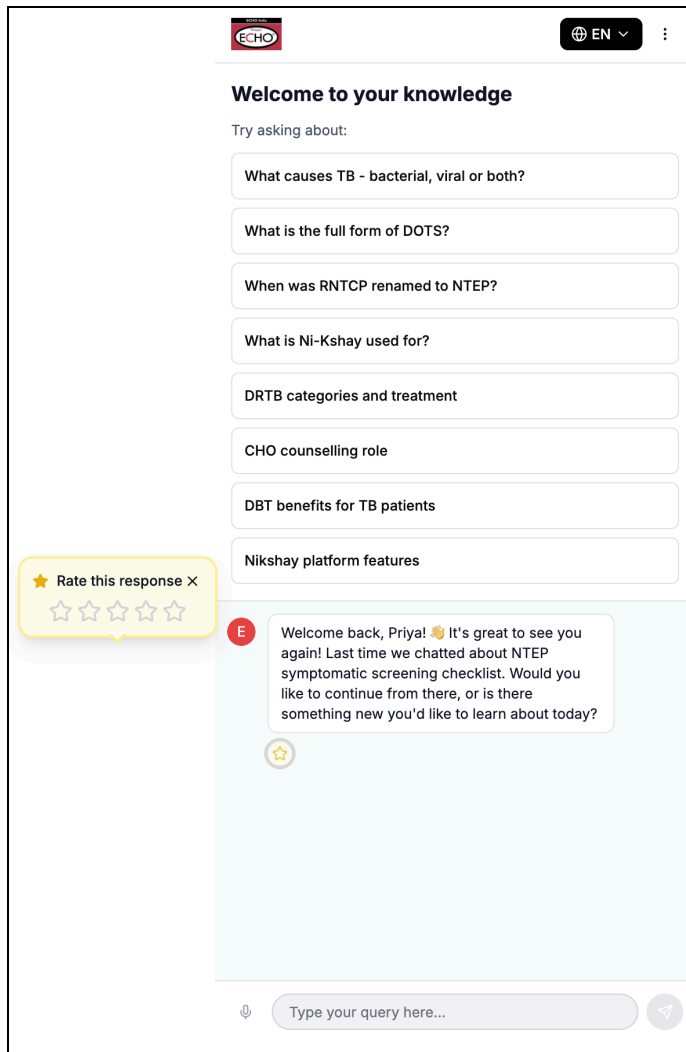
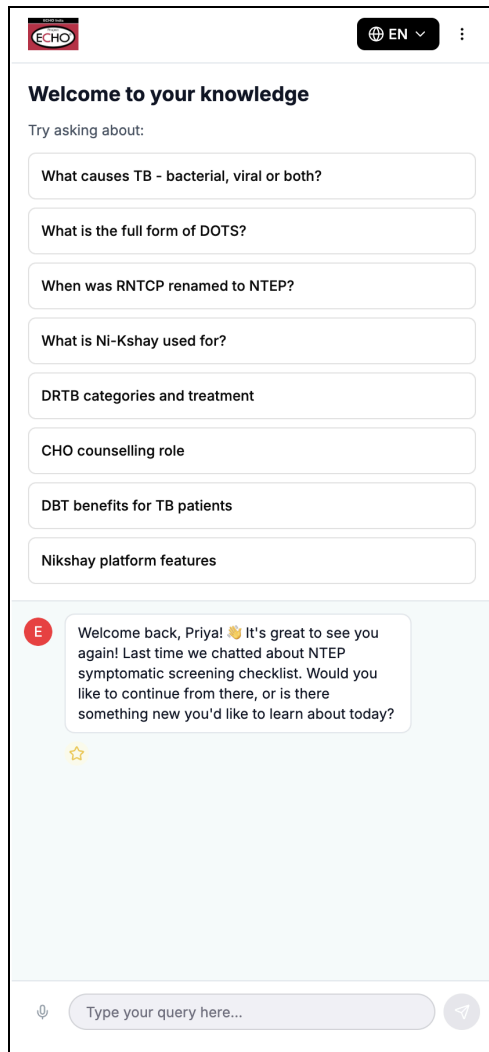
What is the customer opportunity statement?

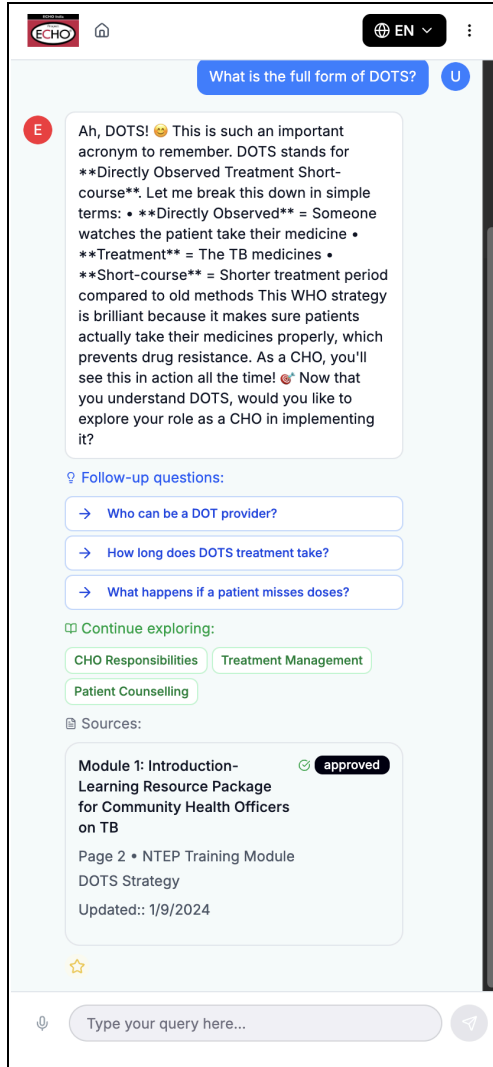
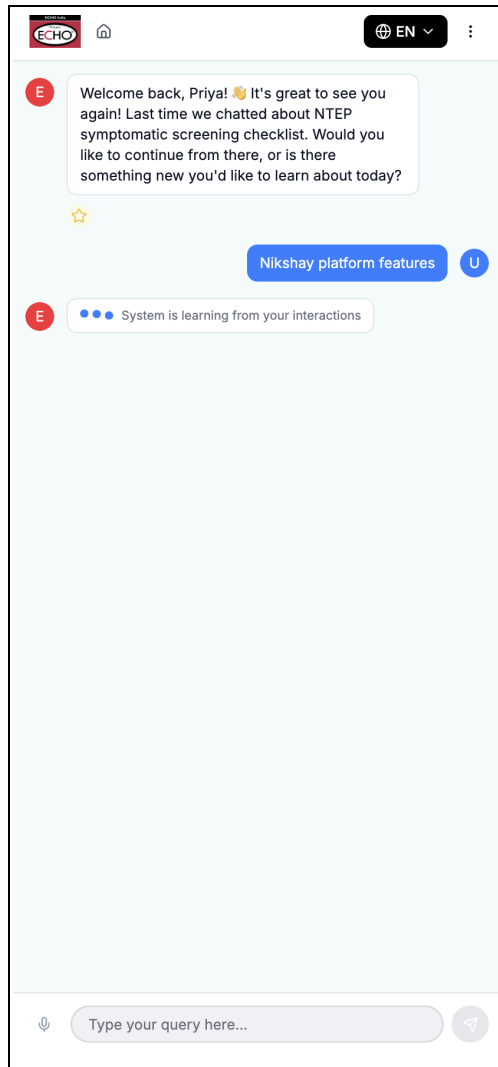
"Community Health Officers, frontline workers, and healthcare professionals in underserved communities need a way to easily access and recall expert-approved knowledge from their training programs because due to heavy workloads, poor connectivity, and limited retention, many find it difficult to effectively apply what they have learned in critical field situations, leading to suboptimal patient care and missed opportunities for impact."

"At the same time, program experts and organizations need a way to ensure their approved knowledge reaches learners effectively and can be accessed on-demand because traditional training materials are often scattered, hard to navigate, and not accessible in real-time when frontline workers need them most during critical decision-making moments."

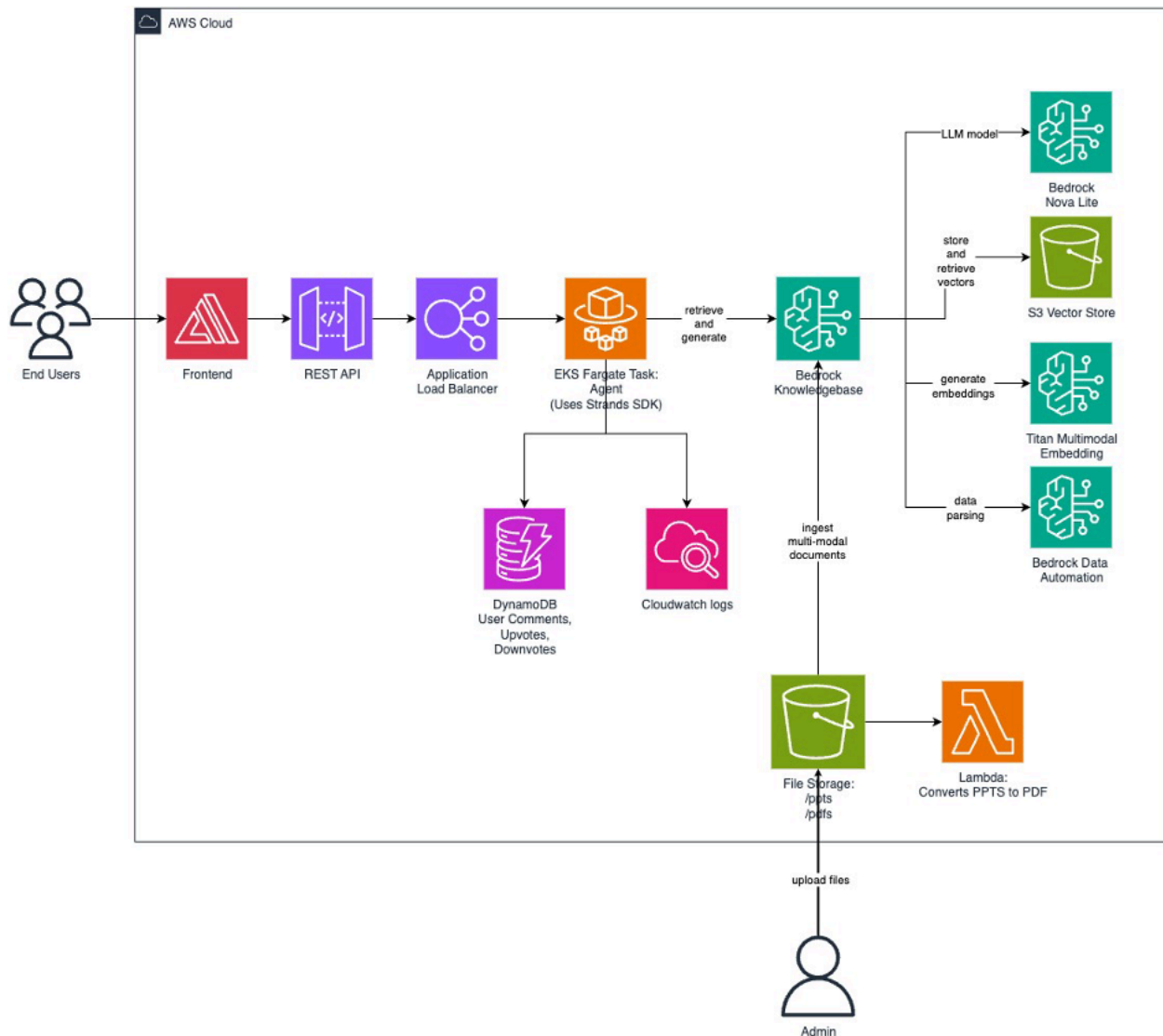
Key Chatbot Features for Mobile-First Experience:

- **Five-star rating system** for comprehensive feedback on response quality and usefulness
- **Frequently Asked Questions (FAQ)** section with quick-access buttons for common queries
- **Suggested question prompts** based on the user's program and recent session content
- **Citation and source references** showing exactly where information comes from in approved documents
- **Mobile-optimized interface** designed specifically for low-end Android devices and slow internet connections
- **Offline-friendly design** that works efficiently even with poor connectivity





Architecture Design:



Architecture Description:

1. User Interaction Layer

- End Users interact via a Frontend UI (e.g., React, Angular).
- The frontend communicates with the backend through:
 - Amazon API Gateway which securely exposes the backend REST APIs to the internet.
 - API Gateway routes requests to the REST API service, which is behind an Application Load Balancer (ALB).

2. Application and Task Orchestration

- The REST API forwards user requests to an EKS Fargate Task (Agent).

- This Agent uses the Strands SDK to orchestrate tasks like document retrieval, ranking, and LLM-based generation.
- It interacts with:
 - Bedrock Knowledgebase for information retrieval and generation.
 - CloudWatch Logs for observability and monitoring.
 - DynamoDB to store and retrieve user feedback and response ratings.

3. Document Ingestion Pipeline (Admin Workflow)

- Admin users upload PPT or PDF files into Amazon S3 (File Storage).
- An AWS Lambda function is triggered:
 - Converts PPT files to PDFs to maintain a consistent document format.
- The Bedrock Knowledgebase ingests these multi-modal documents for further processing.

4. AI/ML and Knowledgebase Integration

- The Bedrock Knowledgebase performs:
 - Data parsing using Bedrock Data Automation.
 - Embedding generation using Titan Multimodal Embedding.
 - Stores and retrieves vector embeddings in an S3-based Vector Store.
- LLM responses are generated using Bedrock Nova Lite, which works alongside the knowledge base to serve responses.

5. Storage, Monitoring, and Data Services

- Amazon S3:
 - Stores uploaded files (PDFs/PPTs).
 - Stores vector embeddings used by the knowledge base.
- Amazon CloudWatch: Captures logs from the EKS agent for monitoring.
- Amazon DynamoDB: Manages metadata such as user feedback and response ratings.

Cost Estimation:

Estimation Summary:								
Service		Monthly Cost (USD)	Monthly Cost (INR)	Annual Cost (USD)	Annual Cost (INR)		Upfront Cost (USD)	Upfront Cost (INR)
S3 Standard		9.2	793.96	110.4	9527.52		0	0
Amazon API Gateway		1.47	126.861	17.64	1522.332		0	0
AWS Amplify		10.61553192	916.120405	127.3863831	10993.44486		0	0
AWS Fargate		36.0401	3110.26063	432.4812	37323.12756		0	0
Amazon EKS		73	6299.9	876	75598.8		0	0
DynamoDB provisioned capacity		14.344	1237.8872	172.128	14854.6464		150	12945
Application Load Balancer		17.22499824	1486.517348	206.6999785	17838.20818		0	0
AWS Lambda (PPT to PDF)		0.7120014	61.44572082	8.5440168	737.3486498		0	0
Nova Lite		28.512	2460.5856	342.144	29527.0272		0	0
Titan Multimodal Embedding		35.372	3052.6036	424.464	36631.2432		24.36	2102.268
Bedrock Data Automation		230	19849	2760	238188		13800	1190940
Cloudwatch Logs		1	86.3	12	1035.6		0	0
S3 vector store		4.01807375	346.7597646	48.21688495	4161.117175		10	863
Total:		461.5087053	39828.20127	5538.104464	477938.4152		13984.36	1206850.268
Detailed breakdown:								
Exchange Rate (USD to INR)		86.3						
S3 file storage								
Unit (GB)	Cost per unit (USD)	Cost (USD)	Cost (INR)					
400	0.023	9.2	793.96					
API Gateway								
Requests in thousands	REST API unit multiplier	Cost per unit (USD)	Cost (USD)	Cost (INR)				
420	1000	0.0000035	1.47	126.861				
AWS Amplify								
Build minute cost								
Build minutes	Cost per minute (USD)	Cost (USD)	Cost (INR)					
10	0.01	0.1	8.63					

Data stored cost								
Unit (GB)	Cost per unit (USD)	Cost (USD)	Cost (INR)					
1	0.023	0.023	1.9849					
Data served cost								
Unit (GB)	Cost per unit (USD)	Cost (USD)	Cost (INR)					
50	0.15	7.5	647.25					
SSR request cost								
SSR requests per hour	Hours in month	Cost per request(USD)	Cost (USD)	Cost (INR)				
146	730	0.0000003	0.031974	2.7593562				
SSR compute duration cost								
SSR requests per month	Avg runtime (ms)	Cost per second (USD)	Cost (USD)	Cost (INR)				
106580	500	5.56E-05	2.960557924	255.4961488				
Total Amplify hosting Cost (USD)	Total Amplify hosting Cost (INR)							
10.61553192	916.120405							
AWS Fargate								
Number of tasks per day	Duration (hrs)	Number of tasks per month						
1	24	30.41666667						
per vCPU hours cost								
Cost per hour (USD)	Cost (USD)	Cost (INR)						
0.04048	29.5504	2550.19952						
GB hours cost (2GB)								
Cost per hour (USD)	Cost (USD)	Cost (INR)						
0.004445	6.4897	560.06111						
Fargate monthly cost								
USD	INR							
36.0401	3110.26063							

Amzon EKS (standard)								
Num of Clusters	Cost per hour (USD)	Hours per month	Cost (USD)	Cost (INR)				
1	0.1	730	73	6299.9				
DynamoDB								
Data Storage								
Storage size (GB)	Cost per GB (USD)	Cost (USD)	Cost (INR)					
20	0.25	5	431.5					
Write setting								
WCU hourly cost (USD)	reserved capacity WCUs	Total monthly hours	Cost (USD)	Cost (INR)				
0.000128	100	730	9.344	806.3872				
DynamoDB monthly cost								
USD	INR							
14.344	1237.8872							
Upfront cost								
reserved capacity WCUs	upfront reserved capacity WCU cost (USD)	Cost (USD)	Cost (INR)					
100	1.5	150	12945					
Load Balancer								
Service settings								
Num of balancers	Cost per hour (USD)	Hours per month	Cost (USD)	Cost (INR)				
1	0.0225	730	16.425	1417.4775				
Load Balancer Capacity Units (LCUs)								
Num of balancers	LCUs	LCU cost per hour (USD)	Hours per month	Cost (USD)	Cost (INR)			
1	0.136986	0.008	730	0.79999824	69.03984811			

Avg Input tokens per request	Cost per 1000 input Token (USD)	Cost for input tokens (USD)	Cost for input tokens (INR)					
100	0.00006	2.592	223.6896					
Avg Output tokens per request	Cost per 1000 output Token (USD)	Cost for output tokens (USD)	Cost for output tokens (INR)					
250	0.00024	25.92	2236.896					
		USD	INR					
Total monthly cost		28.512	2460.5856					
Bedrock (Titan Multimodal Embeddings)								
Avg requests per minute	Hours per day at this rate							
10	24							
Query Embeddings								
Avg Input tokens per request	Cost per 1000 input Token (USD)	Cost for input tokens (USD)	Cost for input tokens (INR)					
100	0.0008	34.56	2982.528					
Vector Embeddings								
Upfront (existing 60000 documents)	Note: Uses batch processing							
Num of documents	Avg no. of pages per document	Avg no. of images per document	Avg no. of tokens per page	Cost per 1000 input tokens (USD)	Cost per input image (USD)	Cost (USD)	Cost (INR)	
60000	20	10	50	0.0004	0.00003	24.36	2102.268	
Monthly	Note: Uses on demand , currently don't have actual stats for this, so I am assuming about additional 1000 documents will be uploaded each month							
Num of documents	Avg no. of pages per document	Avg no. of images per document	Avg no. of tokens per page	Cost per 1000 input tokens (USD)	Cost per input image (USD)	Cost (USD)	Cost (INR)	
1000	20	10	50	0.0008	0.00006	0.812	70.0756	
Bedrock Data Automation								
Upfront								
Avg number of pages per document	Number of documents	Cost per page (USD)	Avg num of images per doc	Cost per image (USD)	Estimated cost (USD)	Estimated cost (INR)		
20	60000	0.01	10	0.003	13800	1190940		
Monthly								
Avg number of pages per document	Number of documents	Cost per page (USD)	Avg num of images per doc	Cost per image (USD)	Estimated cost (USD)	Estimated cost (INR)		
20	1000	0.01	10	0.003	230	19849		

S3 vector Store								
Storage (Monthly)								
Cost per GB metadata (USD)	Total metadata (in GB)	Cost (USD)	Cost (INR)					
0.06	50	3	258.9					
PUT charge (Upfront)								
Cost per GB metadata (USD)	Total metadata (in GB)	Cost (USD)	Cost (INR)					
0.2	50	10	863					
PUT charge (Monthly)								
Cost per GB metadata (USD)	Additional metadata (in GB)	Cost (USD)	Cost (INR)					
0.2	1	0.2	17.26					
Query charge (Monthly)								
Note I am assuming one vector per page								
Tier 1 (first 100k vectors)								
Num of vectors	Cost per TB (USD)	Data per avg vector (in Kb)	Num of queries	Cost (USD)	Cost (INR)			
100000	0.004	6	360000	0.804662704	69.4423914			
Tier 2 (leftover vectors)								
Num of vectors	Cost per TB (USD)	Data per avg vector (in Kb)	Num of queries	Cost (USD)	Cost (INR)			
20000	0.002	6	60000	0.013411045	1.15737319			
Cloudwatch Logs								
Standard: Data Ingested								
Data in GB	Cost per GB (USD)	Cost (USD)	Cost (INR)					
2	0.5	1	86.3					